Rejection Sampling: Theory and Implementation

Adam Kurth
PHP 2530 — Bayesian Statistics

Spring 2026

1 Motivation

Many Bayesian posteriors and other distributions of interest lack closed-form samplers. Rejection sampling provides a way to draw exact, independent samples from a target density f by using a simpler proposal density g that we *can* sample from. The only requirements are:

1. We can evaluate $f(x)$ pointwise (up to a normalizing constant).
2. There exists a proposal g and a finite constant $c \geq 1$ such that $c\,g(x) \geq f(x)$ for all x.

2 The Algorithm

Algorithm 1 Rejection Sampling

```
repeat
    Draw Y ~ g                                   ▷ proposal
    Draw U ~ Uniform(0, 1)                       ▷ independent
    if U ≤ f(Y) / (c g(Y)) then                  ▷ accept with this probability
        accept X = Y
    else
        reject Y and repeat
    end if
until a sample is accepted
```

2.1 Why It Works

The key identity is

$$P(\text{accept} \mid Y = y) \;=\; \frac{f(y)}{c\,g(y)},$$

so the marginal probability of acceptance at any step is

$$P(\text{accept}) = \int \frac{f(y)}{c\,g(y)}\, g(y)\, dy = \frac{1}{c} \int f(y)\, dy = \frac{1}{c}.$$

By Bayes' theorem, the density of an accepted value X is

$$f_X(x) = \frac{P(\text{accept} \mid Y = x)\, g(x)}{P(\text{accept})} = \frac{\frac{f(x)}{c\,g(x)}\, g(x)}{1/c} = f(x).$$

Thus accepted draws are distributed exactly as f. The expected number of proposals per accepted sample is c, so the **acceptance rate** is $1/c$. Minimizing c is therefore critical for efficiency.

3 Implementation in R

The companion script `rejection-sampling.r` demonstrates two examples with four figures. We walk through each block below.

3.1 Example 1: Beta(2.7, 6.3) via Uniform(0, 1)

Setup. The target is $f(x) = \text{Beta}(x;\, 2.7,\, 6.3)$ on $[0, 1]$. The proposal is $g(x) = 1$ (the Uniform density). Since $g(x) = 1$, the envelope constant is simply the maximum of f:

$$c = \max_x f(x) = f\left(\frac{a-1}{a+b-2}\right),$$

evaluated at the mode of the Beta distribution.

```
a <- 2.7; b <- 6.3
mode_beta <- (a - 1) / (a + b - 2)
c_beta <- dbeta(mode_beta, a, b)
```

This gives $c \approx 2.67$, so the theoretical acceptance rate is $1/c \approx 37.5\%$.

Sampling. We draw $N = 10{,}000$ proposals $x_i \sim \text{Uniform}(0, 1)$ and $u_i \sim \text{Uniform}(0, 1)$. We accept x_i whenever $u_i \leq f(x_i)/c$:

```
N <- 10000
x_prop <- runif(N)
u <- runif(N)
accept_prob <- dbeta(x_prop, a, b) / c_beta
accepted <- u <= accept_prob
```

Figure 1(a): Envelope and dart-board visualization. Each proposal generates a point $(x_i,\ u_i \cdot c)$ in the rectangle $[0, 1] \times [0, c]$. Points falling under the curve $f(x)$ are accepted (green); those above are rejected (red). This is the "dart throwing" interpretation: we throw darts uniformly at the bounding box $c \cdot g(x)$ and keep only those landing under f.

```
points(x_prop[1:n_show], u[1:n_show] * c_beta,
       pch = 20, cex = 0.35,
       col = ifelse(accepted[1:n_show],
                    rgb(0.2, 0.6, 0.2, 0.5),
                    rgb(0.8, 0.2, 0.2, 0.3)))
```

Figure 1(b): Histogram vs. true density. The histogram of accepted x-values is overlaid with the true $\text{Beta}(2.7, 6.3)$ density, visually confirming the correctness of the sampler.

3.2 Example 2: $N(0,1)$ via t_2

Setup. The target is $f(x) = \phi(x)$, the standard normal density. The proposal $g(x) = t_2(x)$ has heavier tails, guaranteeing that $c \cdot g(x) \geq f(x)$ everywhere. The optimal c is found numerically:

```
ratio_fn <- function(x) dnorm(x) / dt(x, df = 2)
opt <- optimize(ratio_fn, interval = c(-5, 5), maximum = TRUE)
c_norm <- opt$objective
```

This yields $c \approx 1.257$, giving an acceptance rate of $\approx 79.5\%$—much more efficient than the Beta example because the t_2 envelope hugs ϕ closely.

Figure 1(c): Envelope comparison. The shaded region between $c \cdot t_2(x)$ and $\phi(x)$ represents wasted proposals. A tighter envelope (smaller shaded area) means higher efficiency.

Figure 1(d): Histogram validation. As before, the histogram of accepted samples closely follows $\phi(x)$.

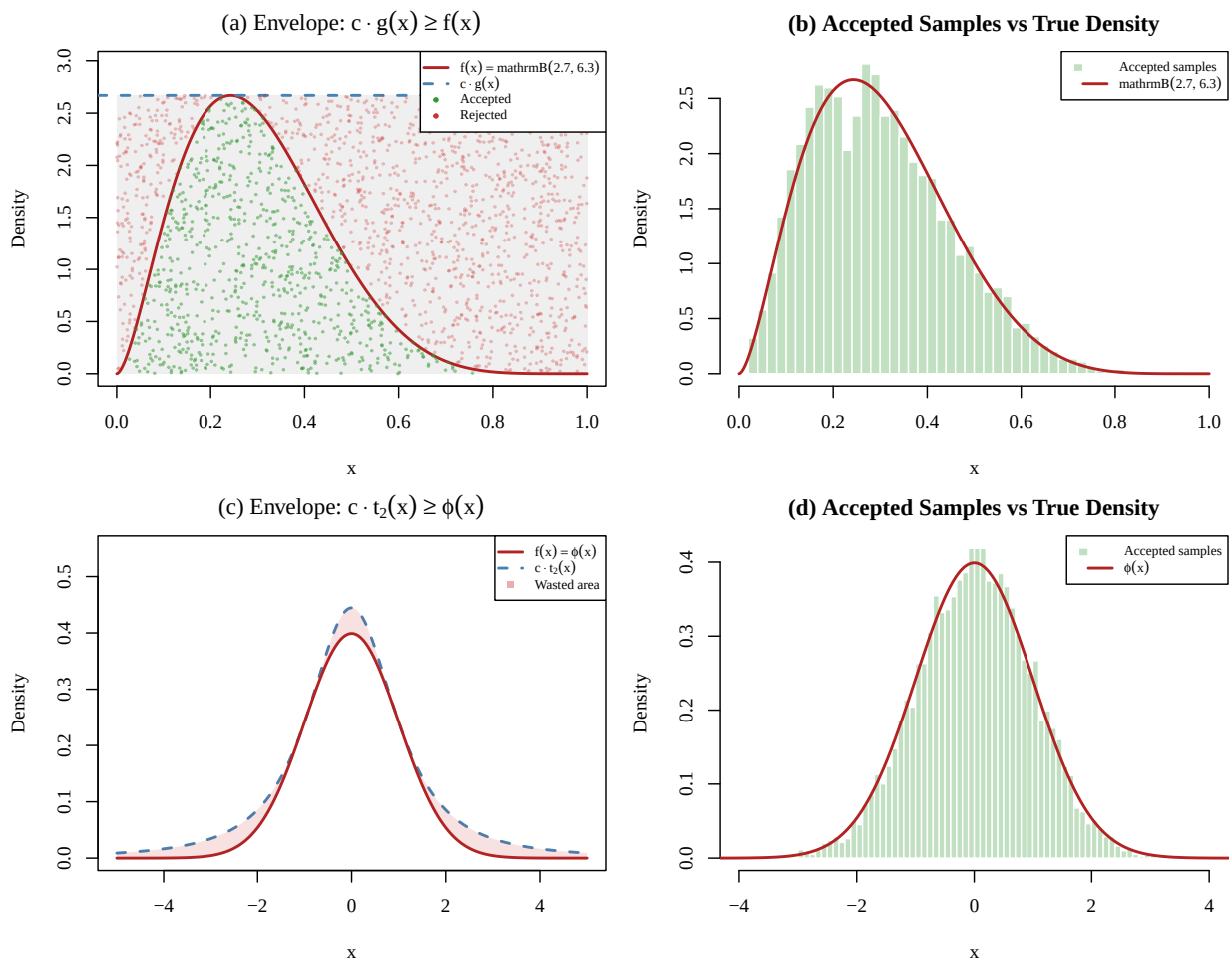

Figure 1: **(a)** Dart-board visualization for the Beta$(2.7, 6.3)$ target with a Uniform$(0,1)$ proposal. Green points fall under $f(x)$ and are accepted; red points are rejected. **(b)** Histogram of accepted samples overlaid with the true Beta density. **(c)** Envelope $c \cdot t_2(x)$ over the standard normal target $\phi(x)$; the shaded region is wasted probability. **(d)** Histogram of accepted samples vs. $\phi(x)$.

3.3 Figure 2: Efficiency vs. Envelope Constant

The acceptance rate is exactly $1/c$. Figure 2 plots this relationship for a range of c values above the optimal c^*, illustrating why minimizing c matters. Choosing c even $2\times$ too large cuts efficiency in half.

```
c_vals <- seq(c_beta, c_beta * 3, length.out = 30)
eff <- 1 / c_vals
plot(c_vals, eff * 100, type = "b", ...)
```

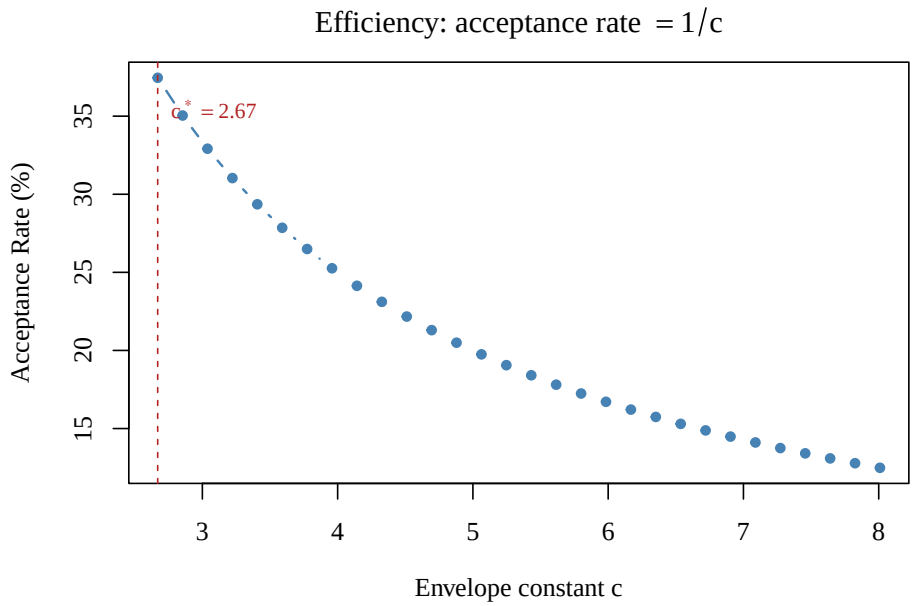

Figure 2: Acceptance rate $1/c$ as a function of the envelope constant c. The dashed vertical line marks the optimal $c^* \approx 2.67$ for the Beta$(2.7, 6.3)$/Uniform example. Larger c wastes more proposals.

3.4 Figure 3: Step-by-Step Visualization

The final figure isolates six individual iterations of Algorithm 1. For each step:

1. A proposal x is drawn from g (the x-coordinate).
2. A uniform u is drawn; the point $(x, u \cdot c)$ is plotted.
3. If the point falls *under* $f(x)$, the panel is labeled "Accept"; otherwise "Reject."
4. An arrow from the x-axis to the point illustrates the "dart throw."

Previous points accumulate across panels, building intuition for how the sample grows.

```
for (k in 1:steps) {
  # ... draw envelope and target ...
  pt_col <- ifelse(acc_seq[k], "green4", "red3")
  points(x_seq[k], u_seq[k] * c_beta,
         pch = 8, cex = 2, col = pt_col, lwd = 2)
```

```
  arrows(x_seq[k], 0, x_seq[k], u_seq[k] * c_beta,
         length = 0.08, col = pt_col, lty = 3)
}
```

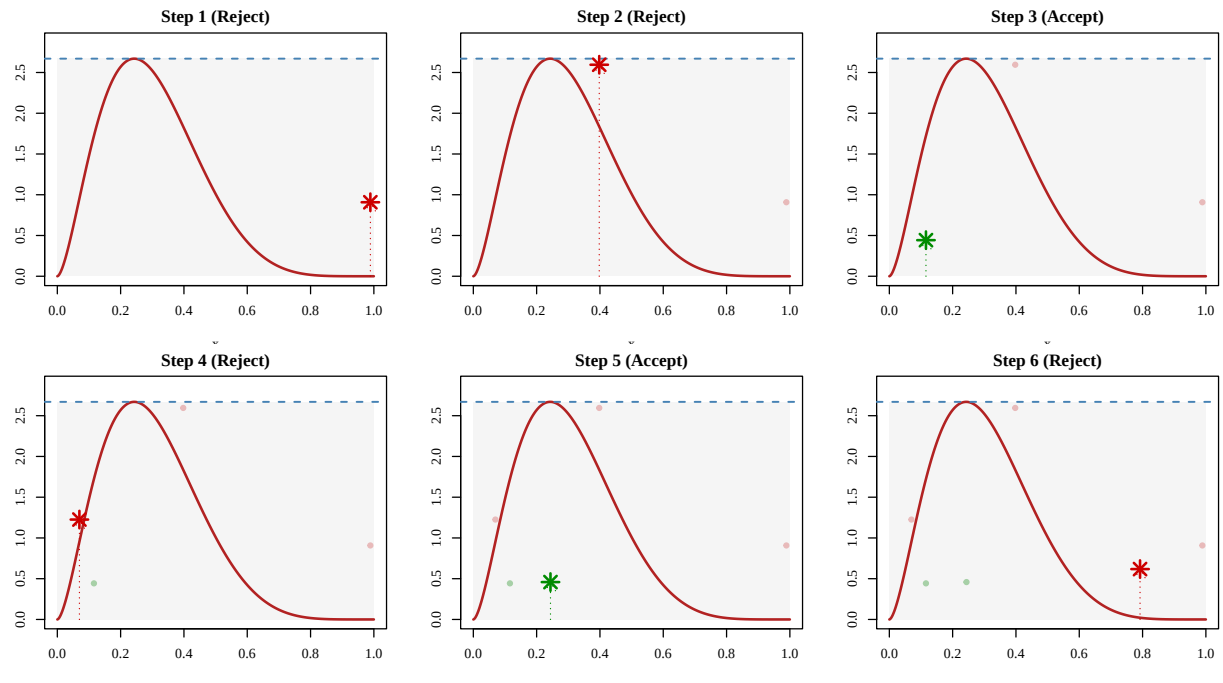

Figure 3: Six sequential iterations of rejection sampling for Beta$(2.7, 6.3)$. Each panel adds one proposal: a star marks the current draw, with an arrow from the x-axis showing its height $u \cdot c$. Green = accepted (under the curve), red = rejected (above). Previous points accumulate across panels.

4 The 2D Picture: Contour Visualization

To build geometric intuition, we apply rejection sampling to a two-dimensional target $p(\mathbf{x}) = N(\mathbf{0}, I_2)$ using a uniform proposal on $[-3, 3]^2$. In 2D the "dart-board" is a square, and accepted points cluster inside the circular contours of the bivariate normal.

Figure 4(a): 2D accept/reject. Green points falling inside the high-density contours are accepted; red points in the corners of the bounding box are rejected. The wasted area in the corners is already substantial in two dimensions.

Figure 4(b): Curse of dimensionality. Panel (b) plots the theoretical acceptance rate $1/c = (\sqrt{2\pi}/6)^d$ as a function of dimension d for the same $N(\mathbf{0}, I_d)$/Uniform setup. At $d = 1$ the acceptance rate is $\approx 41.7\%$; by $d = 10$ it is below 0.02%. This exponential decay is why rejection sampling is impractical beyond a handful of dimensions.

```
# Theoretical acceptance rate for N(0,I_d) with Uniform[-3,3]^d
acc_rates[di] <- (sqrt(2 * pi) / 6)^d
```

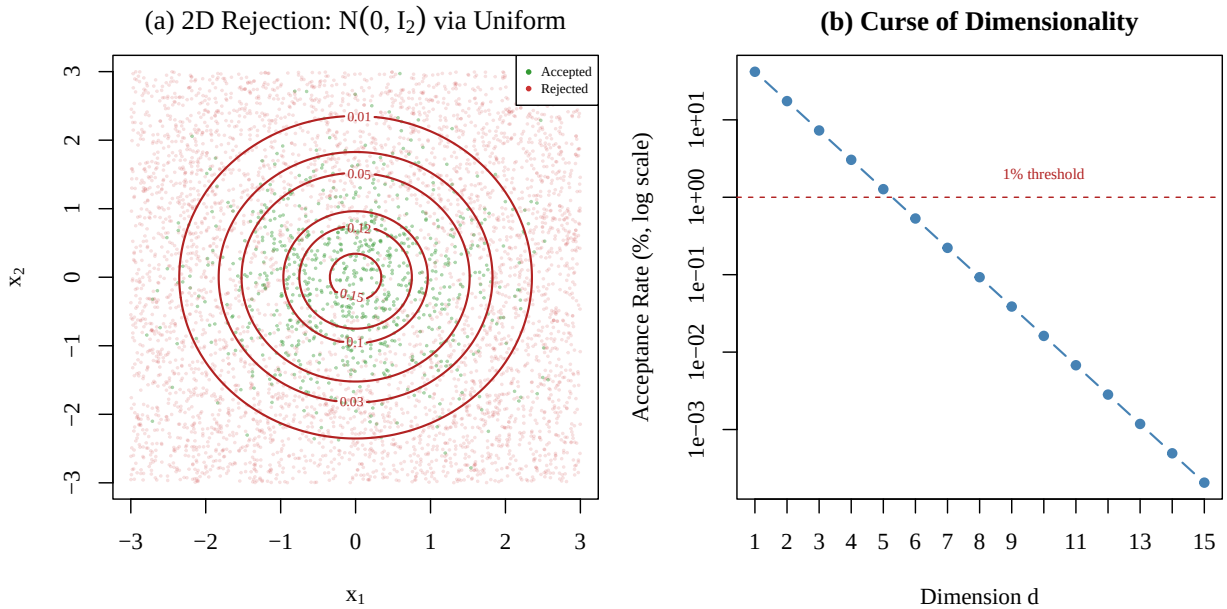

Figure 4: **(a)** Rejection sampling on a bivariate $N(\mathbf{0}, I_2)$ target with a uniform proposal on $[-3, 3]^2$. Accepted samples (green) concentrate inside the density contours; rejected samples (red) fill the wasted corners. **(b)** Acceptance rate decays exponentially with dimension: at $d = 10$ it is below 0.02%.

5 Where Rejection Sampling Fails

5.1 Failure Mode: No Finite Envelope Exists

Rejection sampling requires a finite constant c such that $c \cdot g(x) \geq f(x)$ for *all* x. This condition is violated when the target f has **heavier tails** than the proposal g.

Example. Let the target be $f(x) = \text{Cauchy}(0, 1)$ and the proposal be $g(x) = N(0, 1)$. As $|x| \to \infty$,

$$\frac{f(x)}{g(x)} = \frac{(1/\pi)(1 + x^2)^{-1}}{(2\pi)^{-1/2}\, e^{-x^2/2}} \sim \frac{\sqrt{2\pi}}{\pi} \frac{e^{x^2/2}}{1 + x^2} \to \infty.$$

No finite c can contain this ratio. Figure 5(a) shows the Cauchy density poking above any scaled normal envelope in the tails, and Figure 5(b) confirms the ratio $f(x)/g(x)$ diverges.

Fix. Reverse the roles: use $g = \text{Cauchy}$ to sample from $f = N(0, 1)$. More generally, the proposal must always have **at least as heavy tails** as the target. If no such g is available (e.g., the target's tail behavior is unknown), rejection sampling cannot be applied and one must turn to MCMC or importance sampling.

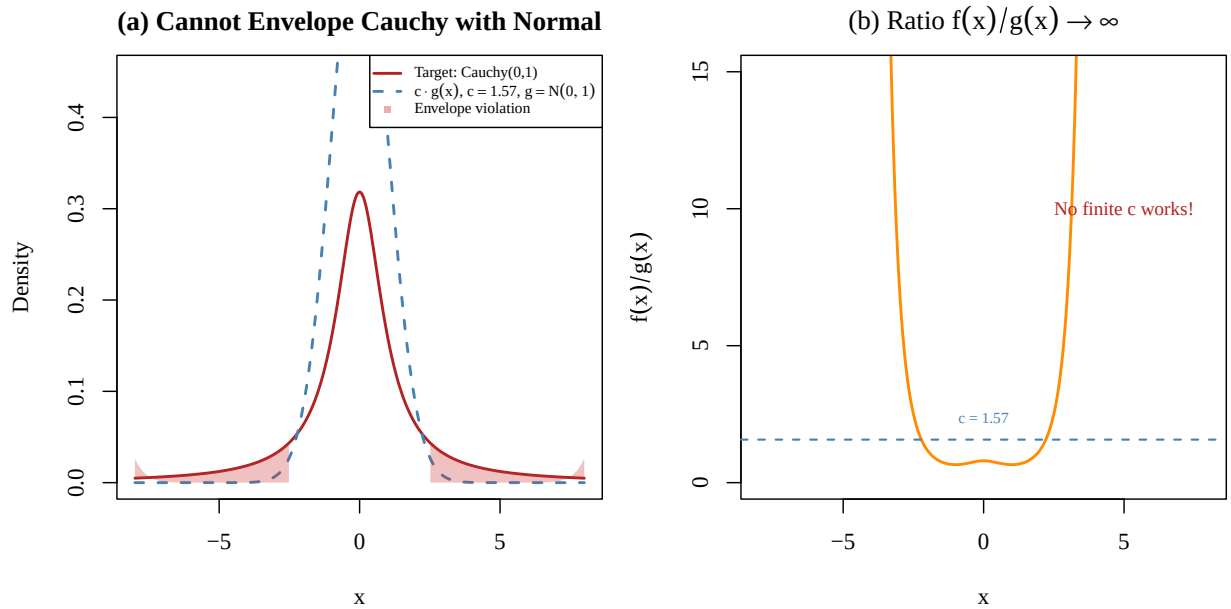

Figure 5: **(a)** Attempting to envelope the Cauchy target (red) with a scaled normal (blue dashed). The shaded regions mark where the envelope is violated—no finite c fixes this. **(b)** The ratio $f(x)/g(x) = \text{Cauchy}(x)/\phi(x)$ diverges in the tails, confirming the impossibility.

6 Key Takeaways

- Rejection sampling produces **exact, independent** draws from f—no burn-in or autocorrelation.
- Efficiency is governed entirely by the constant c: acceptance rate $= 1/c$.
- In **low dimensions**, a good proposal g that closely matches f makes rejection sampling practical.
- In **high dimensions**, the ratio of the volume under f to the volume under $c \cdot g$ shrinks exponentially, making acceptance rates vanishingly small. This motivates MCMC methods (Metropolis–Hastings, Gibbs) that sacrifice independence for guaranteed samples.
- The Uniform proposal in Example 1 ($1/c \approx 37.5\%$) is far less efficient than the t_2 proposal in Example 2 ($1/c \approx 79.5\%$), illustrating the importance of matching the proposal shape to the target.
- Rejection sampling **cannot** be used when the proposal has lighter tails than the target (e.g., normal proposal for a Cauchy target)—no finite c exists.